

EQ Credit: Interior Lighting

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1 point)

INTENT

To promote occupants' productivity, comfort, and well-being by providing high-quality lighting.

REQUIREMENTS

NC, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY

Meet 1 strategy for 1 point. Meet 3 strategies total for 2 points.

1. Glare Control

For all regularly occupied spaces, meet one of the following requirements:

- ▶ Use light fixtures with a luminance of less than 7,000 candela per square meter (cd/m)² between 45 and 90 degrees from nadir.

OR

- ▶ Achieve a Unified Glare Rating (UGR) rating of <19 using software modelling calculations of the designed lighting.

Exceptions include wallwash fixtures properly aimed at walls, as specified by manufacturer's data, indirect uplighting fixtures, provided there is no view down into these uplights from a regularly occupied space above, and any other specific applications (i.e. adjustable fixtures).

2. Color Rendering

For all regularly occupied spaces meet one of the following requirements:

- ▶ Use light sources that have a Color Rendering Index (CRI) of at least 90.
- ▶ Use light sources that have a Color Fidelity Index greater than or equal to 78 and a gamut index between 97 and 110, determined in accordance with Illuminating Engineering Society (IES) TM-30.

3. Lighting Control

Provide dimmable or multilevel lighting for 90% of all regularly occupied spaces.

4. Surface Reflectivity

For at least 90% regularly occupied spaces, use interior finishes with a surface reflectance greater or equal to 80% for ceilings and 55% for walls. If included in the project scope, use furniture finishes with a surface reflectance greater or equal to 45% for work surfaces and 50% for movable partitions.

HEALTHCARE

Provide dimmable or multilevel lighting for 90% of all regularly occupied spaces in staff areas.

For at least 75% of patient sleeping rooms, provide lighting controls that are readily accessible from the patient's bed. In patient rooms with more than one patient, the controls must be individual lighting controls. Exceptions include in-patient critical care, pediatric, and psychiatric patient rooms.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The interior lighting credit has been overhauled to prioritize strategies identified by industry lighting experts as most critical and impactful for new construction LEED projects.

Lighting quality strategies for lamp life, direct overhead lighting, and surface illuminance ratio have been removed.

The threshold for light fixture luminance has been decreased from 2,500 candela per square meter (cd/m²) to 7,000 cd/m². An alternative metric Unified Glare Rating has been added as a new option.

The threshold for Color rendering index has been increased from CRI of 80 to CRI of 90. An alternative metric Color Fidelity Index has been added as a new option.

The previous option for lighting control has been simplified to a single requirement for dimmable or multilevel lighting in regularly occupied spaces. Projects no longer have to differentiate between individual and shared multi-occupant spaces. All lighting controls can be shared (individual lighting controls are not required).

The surface reflectance requirements have been combined, requirements for floors are removed and the thresholds have been decreased from 85% for ceilings and 60% for walls to 80% for ceilings and 55% for walls.

Step-by-Step Guidance

Step 1. Establish Lighting Needs

Work with the owner to understand occupant's lighting needs and desires.

- ▶ Document the types of tasks that will occur in each space and the tools or equipment that occupants use regularly, and determine appropriate light levels for tasks.
- ▶ Identify the level of control that occupants should have and the characteristics of the occupant population.
- ▶ Select measures that are appropriate for the project. Some options require attention during early design phases because achievement depends on luminaire selection and configuration and architectural surface specifications.

Step 2. For Projects Pursuing Glare Control

- ▶ For the light fixtures, review luminaire cutsheets, Illuminating Engineering Society photometric files, or other documentation to identify luminance between 45 degrees and 90 degrees from nadir and select products that meet the credit requirements.

- ▶ Alternatively, select fixtures so that the spaces meet the UGR rating requirements.
- ▶ Exceptions include wallwash fixtures properly aimed at walls, as specified by manufacturer's data, indirect uplighting fixtures, provided there is no view down into these uplights from a regularly occupied space above, and any other specific applications (i.e. adjustable fixtures).
- ▶ Exclude shipping and receiving, warehouse, and distribution centers where projects need to meet IES recommended foot-candles while maintaining clearance for forklifts and other tall equipment.

Step 3. For Projects Pursuing Color Rendering

Specify all light sources to meet the credit requirements for color rendering index or color fidelity index.

- ▶ Exceptions include lamps or fixtures specifically designed to provide colored lighting for effect, site lighting, and lamps or fixtures designed for some other special use.
- ▶ For the light sources, determine the CRI, not to be confused with correlated color temperature (CCT) which refers to the spectrum of warm to cool. A light source can have a high or low CRI regardless of CCT.

Step 4. For Projects Pursuing Lighting Control

Design lighting to meet the credit requirements for lighting control.

- ▶ Identify all regularly occupied spaces in the project.
- ▶ Design lighting controls for each of these spaces to meet the credit requirements.

Step 5. For Projects Pursuing Surface Reflectivity

Select or specify high-reflective finish materials. Before construction begins, review manufacturer's cutsheets to identify reflectance typically expressed as a fraction or percentage LR (light reflectance) or LRV (light reflectance value). If manufacturers' data do not include reflectance, measure the reflectance of product samples (before construction) or the installed product (post-construction) using the methodology described in IES Lighting Handbook, or use reflectance charts such as Lighting Guide 11, Surface Reflectance and Color.

For work surfaces and movable partitions compliance is limited to the opaque surfaces of the partition; transparent or partially transparent surfaces are excluded.

If each surface is unable to comply with the reflectance requirement a weighted average surface reflectance calculation can be performed.

Step 6. Confirm Compliance

Compile documentation that confirms compliance with the selected strategies.

Further Explanation

Required Documentation

- ▶ Table of regularly occupied spaces and associated lighting details
- ▶ Lighting details, including manufacturer and model, results of estimations, or in situ or laboratory photometric tests
- ▶ Surface Reflectivity:
 - List of ceiling and wall surfaces, work surfaces (if in project scopes), movable partitions (if in project scope), and their associated surface reflectance values

- Average surface reflectance calculations (if needed)

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to promote occupant comfort and wellbeing by providing high quality lighting in form of designing for lighting controls as well as by providing enhanced lighting quality measures in newly constructed buildings can contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.